

the fraction of this income she saves at the end of each of the thirty years. Savings are deposited into an investment portfolio, which is allocated to an annually rebalanced 50/50 portfolio of the S&P 500 and shorter-term bills. Retirement begins at the start of the thirty-first year and is assumed to last for thirty years. Altogether, the accumulation and decumulation life cycle is sixty years.

Withdrawals are made at the beginning of each year during retirement. The underlying 50/50 asset allocation remains the same during retirement, as does the annual rebalancing assumption. Withdrawal amounts are defined as a replacement rate from final preretirement salary. I assume that the baseline individual wishes to replace 50 percent of her final salary with withdrawals from her accumulated wealth. This 50 percent is more than it may initially seem, as it is only the part from retirement savings. Social Security benefits and any other income sources will be added on top of this. In addition, after retiring, she no longer has to save for retirement or contribute to Social Security, which increases the replacement rate with respect to what could have been spent before retirement. Withdrawal amounts are then adjusted each year for the previous year's inflation. A particular savings rate was successful if it provided enough wealth at retirement to sustain thirty years of withdrawals without having the account balance fall below zero. Actual wealth accumulations and withdrawal rates may vary substantially for different hypothetical individuals from different starting points in history.

Using Robert Shiller's data, I can consider eighty-seven different thirty-year careers followed by thirty-year retirements for retirement dates from 1901 to 1987. In Exhibit 5.8, we saw the historical sustainable spending rates using this data. Exhibit 5.9 shows these outcomes again, along with historical minimum required savings rates to reach a traditional wealth accumulation goal. The savings rates turn Bengen's SAFEMAX calculations on their head in order to calculate a savings rate in isolation from the following retirement period. I calculate the minimum necessary fixed savings rate required over a thirty-year career to achieve a wealth accumulation target at the retirement date. I consider a person wishing to replace 50 percent of her final salary with withdrawals from her accumulated wealth. She wishes to play it safe and hopes to save enough that her desired expenditures represent a 4 percent withdrawal rate from her accumulated wealth. Therefore, she must